A Reusable Agentic Platform for Auditable, Sentiment-Aware Dynamic Asset Allocation

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Project Proposal

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1. Motivation

Banks and asset managers increasingly use AI to support information-heavy work, but financial decisions still require governance, documentation, risk control, and human oversight [1], [2], [3]. This is especially relevant for dynamic stock/bond allocation, since the high-level split between equities and bonds drives much of a portfolio's risk and return [4].

Traditional quantitative models can process valuations, momentum, macro data, rates, and risk indicators, but they usually do not incorporate text-based market sentiment in an auditable way. Generic LLM assistants have the opposite weakness: they can summarize narratives, but they do not provide deterministic analytics, point-in-time traceability, or a controlled link to an asset manager's existing models.

Existing published work already covers financial LLM platforms, LLM-based trading agents, financial report-generation systems, and broader agentic asset-allocation or news-to-portfolio pipelines [5], [6], [7], [8], [9], [10]. However, to the best of our knowledge, this literature has not yet addressed the narrower workflow targeted here: a non-trading, point-in-time, auditable stock/bond allocation memo system that combines deterministic quantitative and sentiment evidence with a constrained LLM report-generation layer.

This project proposes a controlled two-stage system for an asset-management partner. First, numerical and sentiment inputs are converted into a timestamped evidence summary. Second, an LLM uses that summary to generate an investment report and a machine-readable stock/bond allocation decision. The system does not trade, execute allocation changes, or replace the investment committee; it supports the people making the decision.

2. Applied Research Question

> To what extent can an LLM-based report-generation layer, operating only on point-in-time quantitative, macro-financial, and sentiment evidence, support dynamic stock-bond allocation decisions relative to naive 50/50 [11] and conventional 60/40 benchmarks?

This is an applied question, not a purely academic one. The point is to build and evaluate a working prototype, and to see whether agentic AI can be useful, controllable, and inspectable in a realistic allocation workflow the partner actually cares about.

3. Project Objectives

3.1. Objective 1: Build a controlled two-stage evidence-to-report pipeline

Develop a reproducible pipeline in which deterministic tools first generate a structured evidence pack, and the LLM then transforms that evidence into a standardized allocation report. The LLM must not freely browse, trade, or invent unsupported data.

Success criterion: For every rebalancing date, the system produces:

- a structured evidence pack,
- a human-readable investment report,
- a machine-readable allocation decision,
- a record of the model, prompt, data sources, and generation settings.

3.2. Objective 2: Evaluate allocation recommendations against 50/50 and 60/40 benchmarks

Compare the system's recommended stock/bond allocations against two simple baselines: equal-weight 50/50 and conventional 60/40. The 50/50 baseline is the two-asset version of the naive 1/N benchmark used in the portfolio-choice literature [11]. The evaluation should focus on realized out-of-sample portfolio outcomes after each report date. To isolate the LLM's contribution from the evidence pack itself, the system is also compared against a rule-based allocator that consumes the same evidence pack but applies a fixed decision rule instead of the LLM.

Success criterion: The system is evaluated using a predefined set of performance metrics, including return, volatility, Sharpe ratio, drawdown, turnover, and transaction-cost-adjusted performance.

3.3. Objective 3: Assess report quality, auditability, and leakage control

Evaluate whether the generated reports are factually grounded in the evidence pack, internally consistent with the allocation recommendation, and protected against temporal leakage.

Success criterion: Each report can be audited against the evidence pack, and no material factual claim should rely on information unavailable at the decision date.

4. Proposed System

The prototype follows a two-stage workflow. First, numerical inputs from the quant models and text-based inputs from sentiment sources are converted into a structured evidence summary. Second, an LLM uses that summary to generate the investment report and the machine-readable allocation decision. Appendix Section 8 shows an illustrative example of the report format this workflow is meant to produce.

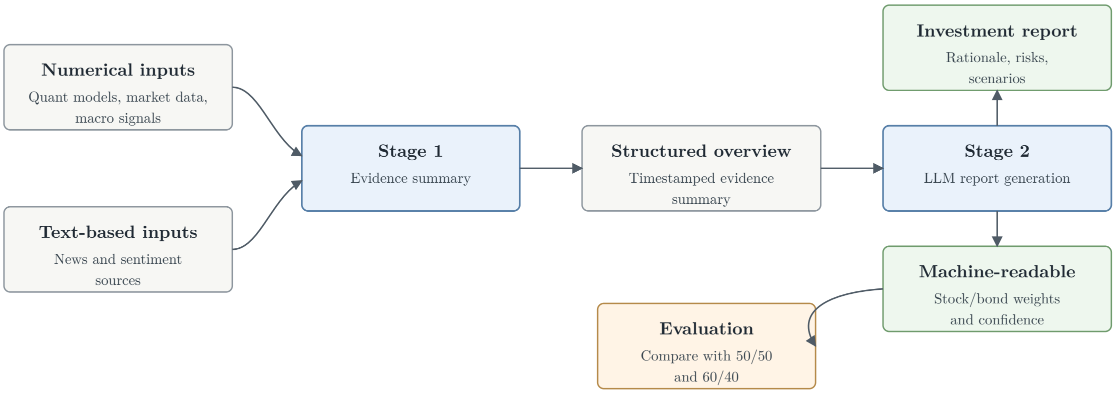

Figure 1: High-level two-stage workflow for the allocation report prototype.

5. Expected Deliverables

The project will deliver a working prototype and evaluation package for the two-stage workflow shown in Figure 1. The deliverables are:

- the two-stage allocation-report pipeline;
- the structured evidence-summary format;
- the generated investment-report template, illustrated in Appendix Section 8;
- the machine-readable allocation-decision format;
- the audit log covering data sources, prompts, model settings, generated outputs, and run metadata;
- the evaluation harness comparing the recommendations against 50/50 and 60/40 baselines, plus a rule-based, no-LLM baseline using the same evidence pack
- the final written report describing the architecture, implementation, evaluation results, limitations, and possible extensions.

6. Evaluation Plan

The evaluation follows an out-of-sample backtesting design, following the portfolio-choice benchmarking logic used by DeMiguel, Garlappi, and Uppal [11]. The implementation model is chosen so that its knowledge cut-off date is known. The backtest period then starts strictly after that cut-off date, so the model cannot rely on memorized information about the evaluated market period. At each rebalancing date, the system may use only information that would have been available at that date. The numerical inputs, sentiment input, evidence summary, prompt configuration, generated report, and allocation decision are all fixed before the next-period returns are observed. This prevents the LLM or the evaluation code from using information from the future.

For each monthly or quarterly rebalancing date t, the pipeline produces a recommended stock/bond allocation w_t. This allocation is then held over the next evaluation interval $(t, t+1]$. The realized portfolio return of the system is compared with two static benchmark portfolios over the same interval: a 50/50 stock/bond allocation and a 60/40 stock/bond allocation. The interval-level benchmark-relative return differences are recorded as:

$$\Delta_{50/50,t+1} = R_{\text{dynamic},t+1} - R_{50/50,t+1}$$
$$\Delta_{60/40,t+1} = R_{\text{dynamic},t+1} - R_{60/40,t+1}$$

With monthly or quarterly rebalancing, a single backtest period yields on the order of ten to thirty allocation decisions, too few to establish statistical significance. The performance comparisons are therefore treated as an illustrative check of the system's behavior rather than as a claim of outperformance.

The report-generation layer is evaluated separately from the economic results. For a sample of generated reports, each material claim is checked against the evidence summary and its source metadata. The evaluation records whether the report is grounded in the available evidence, internally consistent with the machine-readable allocation decision, and free from temporal leakage.

The final assessment therefore has two parts: whether the allocation recommendations are competitive with the 50/50 and 60/40 baselines out of sample, and whether the reports are auditable, timely, and useful as decision-support documents.

7. Scope and Limitations

The project is deliberately limited to one decision-support workflow: the stock/bond allocation report. It covers dynamic allocation (monthly or quarterly) recommendations between equities and bonds, based on a controlled evidence summary that combines deterministic quantitative signals with text-based sentiment input. The numerical side includes market risk and return indicators, interest-rate and yield-curve information, and macro-financial indicators where these are available in point-in-time form. The text-based side is limited to one controlled sentiment source, (i.e. Alpha Vantage, or any other sentiment data provider), so that the provenance and timing of the input can be audited [12]. We verify that retrieved sentiment records reflect the information available at each decision date, rather than any later revision, to avoid look-ahead bias.

The LLM evaluation, such as bias and performance, is out of scope. We will focus on one model that meets our technical needs, such as self-hosting versus API calls, among others.

The system generates an investment report and a machine-readable allocation decision, but it does not trade. It does not create orders, execute transactions, select individual securities, or optimize an unconstrained portfolio. The LLM is used only in the report-generation stage and works from the previously prepared evidence summary. It is not allowed to browse freely, read arbitrary news, call discretionary tools, or invent unsupported data. The concrete model can be adapted to partner constraints, including a self-hosted option if privacy or deployment requirements demand it, but model comparison and fine-tuning are outside the first evaluation phase. The model version and generation settings (e.g., temperature zero) are fixed for the duration of the backtest so that results stay reproducible and unaffected by silent model updates.

The recommendations are compared against the 50/50 and 60/40 stock/bond baselines, using economic performance measures and structured report-quality checks. The project does not attempt a broad benchmark study, multi-asset allocation, personalized financial advice, production-grade compliance approval, or live investment operations. The prototype is therefore an auditable committee-support system, not an autonomous investment system.

References

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8. Appendix: Example Generated Allocation Report

The following example is illustrative. It shows the intended shape of the generated report, not an actual investment recommendation.

Investment Committee Allocation Report
Decision date: 2027-03-31
Evaluation horizon: Next monthly rebalancing interval
Universe: Global equities and investment-grade bonds
Current benchmark: 60/40 stock/bond allocation

Recommended allocation

The system recommends a moderate underweight to equities for the next interval:

- stocks: 55%;
- bonds: 45%.

Evidence summary

The numerical evidence points to a less supportive risk environment than in the previous rebalancing window. Equity momentum remains positive but has weakened, valuation indicators are above their long-run median, and realized volatility has increased. Yield-curve and rates data suggest that duration risk remains material, but the bond allocation still improves portfolio stability under the current risk model.

The text-based sentiment input is mixed but slightly risk-off. Market news sentiment is less positive than in the previous window, with recurring references to tighter financial conditions, slower earnings growth, and uncertainty around central-bank communication. No single article or sentiment signal is treated as decisive; the report uses the sentiment input only as supporting evidence alongside the numerical indicators.

Rationale

The recommended 55/45 allocation keeps the portfolio close to the conventional 60/40 benchmark while reducing equity exposure in response to weaker sentiment and higher measured risk. A larger shift is not recommended because the evidence summary does not show a severe deterioration in macro or market conditions, and the equity signal remains positive on some measures.

Main risks

The recommendation may underperform the 60/40 benchmark if equity markets continue to rally or if the risk-off sentiment proves temporary. It may also underperform the 50/50 benchmark if bond yields rise sharply during the holding period. The allocation should therefore be treated as a controlled tilt, not as a high-conviction market-timing signal.

Audit notes

The report was generated from a timestamped evidence summary. The LLM did not browse the web, call additional tools, or use data outside the evidence pack. The stored audit bundle contains the source references, model identifier, prompt version, generation settings, and the machine-readable allocation decision.